Explicit maps along the twist-derived quotient cycle

All eight degree17 quotient-to-quotient maps are constructed and independently verified on the public p827 example. Both coordinates of each generic commuting square agree, and all6584 rational-point/136 kernel-point checks pass. The initial GP script failed strict verification; an explicitly frozen identifier-only correction verifies the unchanged native data.

Recorded result	Exact count
Degree17 maps and generic squares	8
Rational-point checks	6584
Kernel-point checks, including infinity	136
Wrong-sign rejections	6576
Corrupted curve maps rejected	8

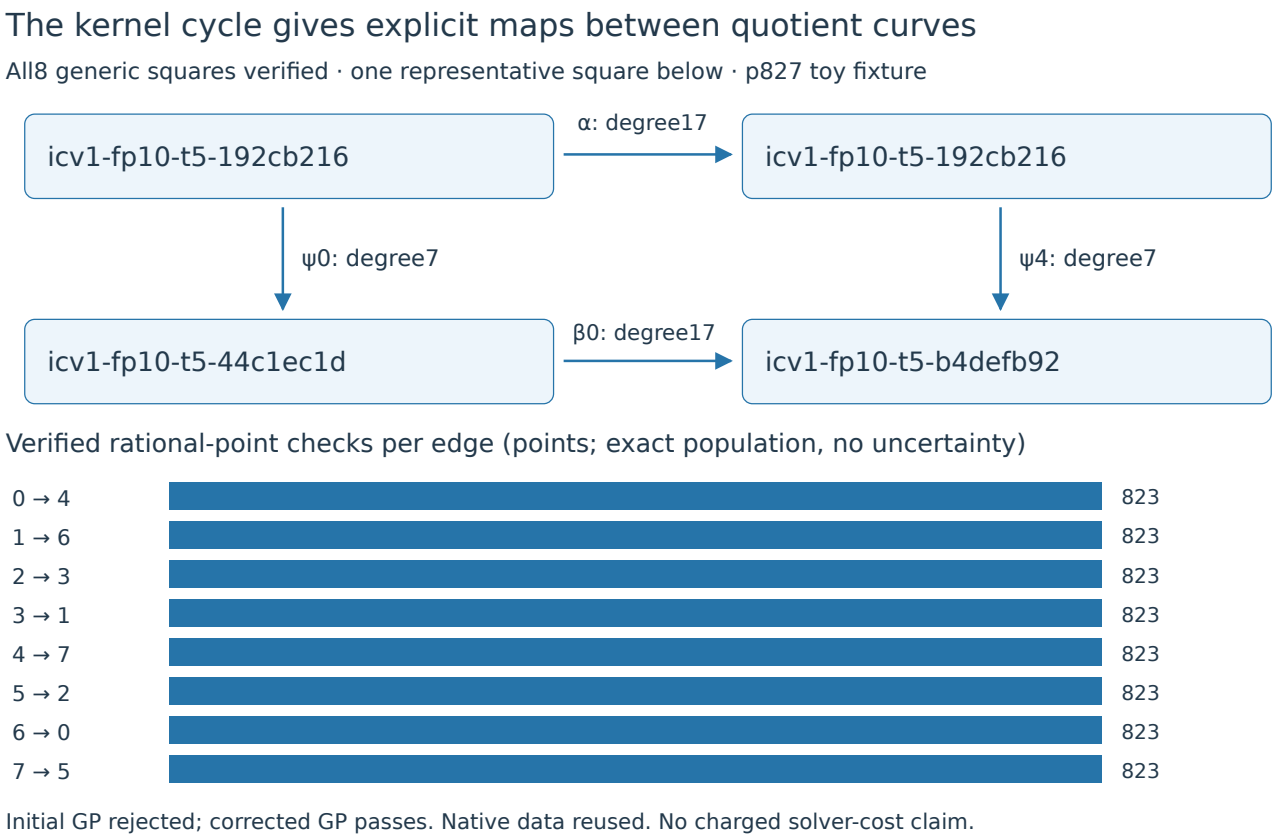


Figure 1: Representative square with full ICV1 model identifiers and all eight point-check populations. Runs: v12 native and v13 corrected GP.

The source $E : y^2 = x^3 + x + 15$ over$\mathbb{F}_{827}$ has order823 and generator $G = (2, 5)$; ICV1 `icv1-fp10-t5-192cb216`. All eight archived degree7 maps $\psi_i : E \to E_i$ and the certified degree17 endomorphism $\alpha = (\pi + [1])/[7]$ are unchanged. The previously verified kernel cycle is $0 \to 4 \to 7 \to 5 \to 2 \to 3 \to 1 \to 6 \to 0$.

For every archived line$i$ and its fixed successor$j$, the new explicit $\beta_i : E_i \to E_j$ satisfies

$$\beta_i \circ \psi_i = \psi_j \circ \alpha.$$

Both sides are verified generically as reduced rational functions; their x-coordinate numerator has degree119. On every rational point$P$, $\beta_i(\psi_i(P)) = [353]\psi_j(P)$. Transporting both generator and

target preserves their DLP scalar; relative to the original next-quotient generator, the factor353 is inverted by415 modulo823.

Construction and independent checks

LetK_{17} be the archived kernel ofα. For each$i$, native Rust forms all17 points of$\psi_i(K_{17})$, checks their order/anti-Frobenius property, and constructs normalized degree17 VéluW_i from eight paired x-roots. The prescribed coordinate scale7 givesβ_i. Exact curve polynomial, degree, coprimality, kernel/poles/infinity and generic square checks pass. All6584 per-point records are retained, rather than a sampled witness list.

PARI/GP independently reconstructsψ_i andα with `ellisogeny`, then uses rational twists ofE_i to build each17-kernel andW_i without sharing the native extension-field engine. It checks all coefficients, generic identities and finite/kernel cases. Negating the output y-coordinate still gives a map to the correct curve but fails the commuting check on all822 nonidentity points per edge; curve membership alone would miss it.

Two Lean declarations, `quotient_descent` and `quotient_descent_unique`, prove the abstract quotient-homomorphism existence and uniqueness from $H \subseteq \alpha^{-1}(K)$. Frozen replay/fresh native transitive axiom audit pass under Lean4.34.1 and pinned mathlib, standard foundations only. The concrete elliptic-curve groups/morphisms, field, kernel inclusion and degree bridge remains unformalized. The fixed-modulus v1 identity-certificate format does not certify these characteristic827 identities; exact native/GP and conditional Lean evidence have their stated scope.

This extends the finite kernel symmetry to executable isogenies between its quotient curves, using standard theory. Complete discovery/evaluation/solver costs and any additional independent relations remain unmeasured. No novelty, cheap degree1 automorphism, IC advantage or NIST applicability is inferred.

Original failure, amendment and reproduction

The original native source is `9e612b707a85ed291f470386126b5b7143628c02`; protocol hash `14c12589c69bb260a9736a8fc332f251c1e18a85c7bba88d981dabb9e54af812`. Native passed. GP used reserved constant `Pi` as a variable, returning0 and the final sentinel with parser/type errors. Strict stderr admission rejected that run. Its script/output/archive and correctly failed export remain intact.

The new script changes only64 literal `Pi→iso_i` occurrences. Source `dc2dddf630c968bc1dcbe6fed5599041b3f8438d`, protocol-v2 hash `eb1830642eb242ea23e616cef59ec8515434f07039249849c76e40a9d9a9dcf0`, freeze/commit and Setec gate precede its single new GP execution. Native strict admission verifies exact replacement bytes and unchanged native data hashes, process0, sole sentinel and only the declared stack-size warning.

Each export has flow planned2/attempted2/process-completed2/interrupted0/ resumed0/excluded0. Original reference verification fails; successor passes. The successor reuses the identical native run. Combined actual total is three processes: native1, failed GP1, corrected GP1; never four independent runs. Both full protocol/file bindings pass. Same author/host, independent implementation; no separate-runner review or statistical inference.

Runs were one thread/CPU0/4GiB/300s each on the managed Xeon8573C KVM host, L0, under the documented EKS access/image fallback. No clocks are admitted, paid resources created or permissions changed. New native controls16/16 and complete standalone runner tests pass; the earlier full-root-suite gap remains.

Exact intermediate ICV1 records and24 generator-bound EC1 model records are in execution-v12, with eight ordered routes. Each endpoint retains $\beta_i(\psi_i(G))$, distinct from the unscaled$\psi_j(G)$; the geometric cycle does not silently equate those generator-bound identities.

Raw v12 archive:75692bytes/nine files, SHA256 `c1d62b826606698abb30fd7bc8210a30e4ff0c83ca26bebaa9dea357a1448f08`. V13 reference archive:1192bytes/six files, SHA256 `6f6158ccf368277e9431a6712d57d6e54cf34ac7bedbefd182e797895db14dae`. Both live in user-confirmed `s3://setec-astronomy-artifacts/store`, with fresh downloads/all-member hashes/identical deterministic

repacks. Execution-v12/v13 READMEs give hydration, exact pins and commands; old waves are unchanged.

Literature read before implementation: MIT18.783 Fall2025 Lecture5, Theorem5.11/function-field uniqueness proof, Corollaries5.10/5.12 and Theorem5.15; Lecture13, Frobenius/endomorphism-ring setting. Also repository GLV/isogeny-loop note and atlas lead. These are existing constructions, not evidence of novelty.

Canonical progress/IC/curve diagrams were checked: no cost row changes. This report adds exact maps, not a full volcano census. Original full research campaigns, rank repair, charged solving and concrete formal bridges stay open.